

Motor Drive Gearbox

OOM-M03-01 / Community Design Reference

DOCUMENT TYPE

COMMUNITY DESIGN REFERENCE

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B

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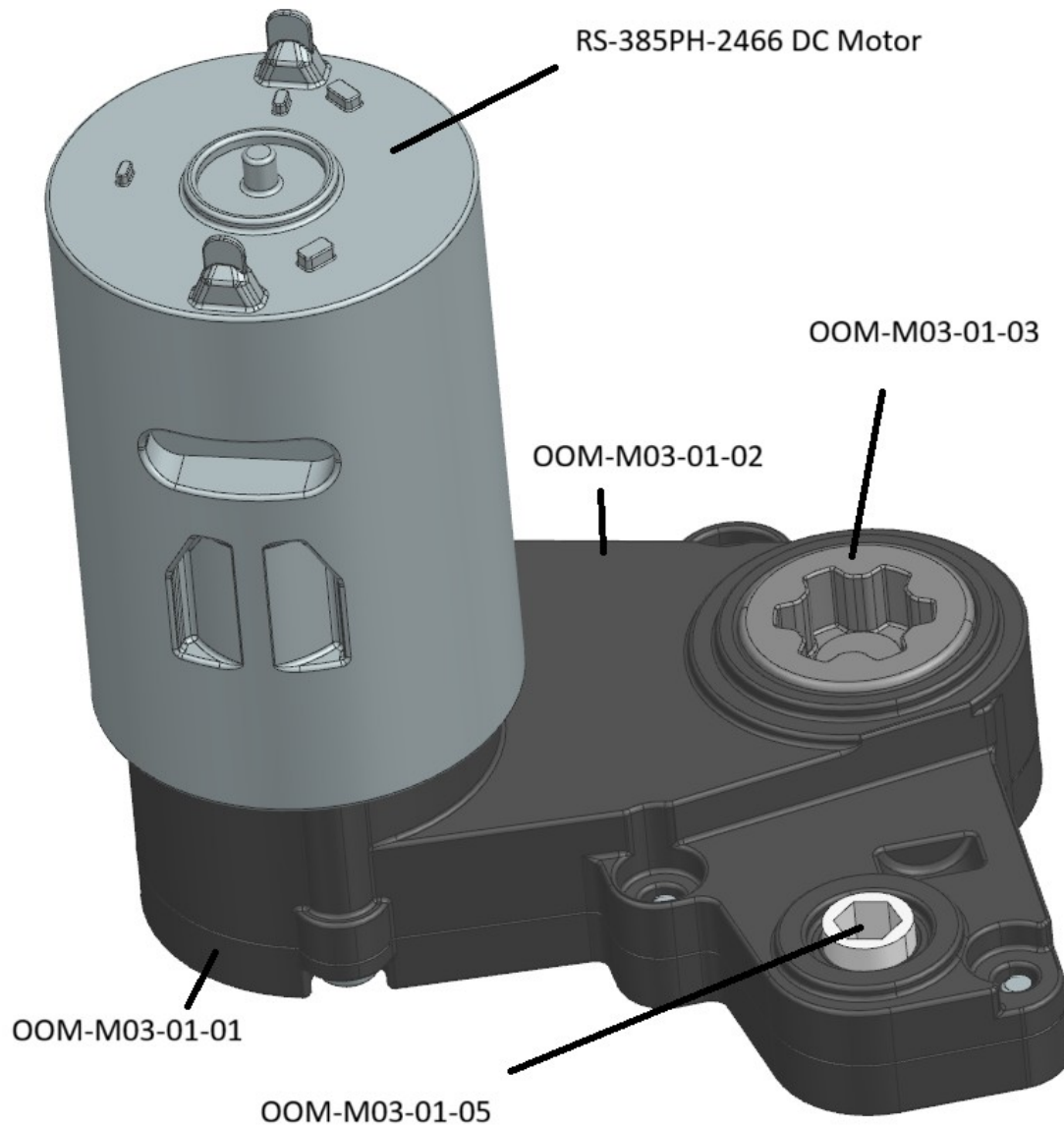


Figure 1. OOM-M03-01 motor-drive gearbox with RS-385PH-2466 motor, housing part codes and external output interface.

Purpose. Explain the OOM-M03-01 motor gearbox architecture, helical input stage, spur transfer train, housing interfaces and shaft/support logic for the OOMWOO One community.

COMMUNITY USE / Use this document together with the native CAD and STEP files. The tables explain the design intent and parameter relationships; the current repository CAD remains the master geometry for manufacturing, remixing or modification.

0 About This Design Reference

A practical engineering companion for understanding, rebuilding and modifying the OOM-M03-01 gearbox module.

OOMWOO One is an open-source robot vacuum project intended to be buildable, documented and hackable. This reference explains the mechanical logic behind the OOM-M03-01 gearbox so a community member can understand the motion path, identify the custom parts, reproduce the gear parameters and preserve the critical housing/shaft relationships when adapting the module.

How to use this document

- Part-code traceability: custom components are identified by their OOM-M03-01-xx project codes shown in the annotated CAD views.
- CAD-first reading: dimensions and formulas here explain the design, while the repository CAD/STEP geometry remains the manufacturing master.
- Kinematic clarity: the first helical reduction stage and the following spur transfer path are separated so the ratio logic is easy to follow.
- Interface-first detail: motor mounting, compound-gear coaxiality, housing journals/posts and the output socket are treated as mechanical interfaces rather than isolated parts.
- Purchased-hardware traceability: bearings, bushings and shaft diameters identified in the annotated housing views are documented in Section 6 and the BOM. Axial lengths, material choices and any unlabelled fit details remain controlled by the current CAD/BOM.

Design basis

ID	Design basis	Defines	Use in this document
B1	OOM-M03-01 CAD / STEP	Gear geometry, housing interfaces and output coupling	Manufacturing master
B2	Annotated CAD figures	Part codes, motor model and assembly orientation	Visual orientation
B3	OOMWOO One CAD / project documentation	Open-hardware context and gearbox sourcing philosophy	Project context
B4	Motor reference data	RS-385PH family and exact RS-385PH-2466 12 V identification	Purchased motor context

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1 Assembly Architecture & Part Numbering

A seven-gear transmission packages a high-ratio helical input stage and a long spur-gear transfer path inside a two-piece housing.

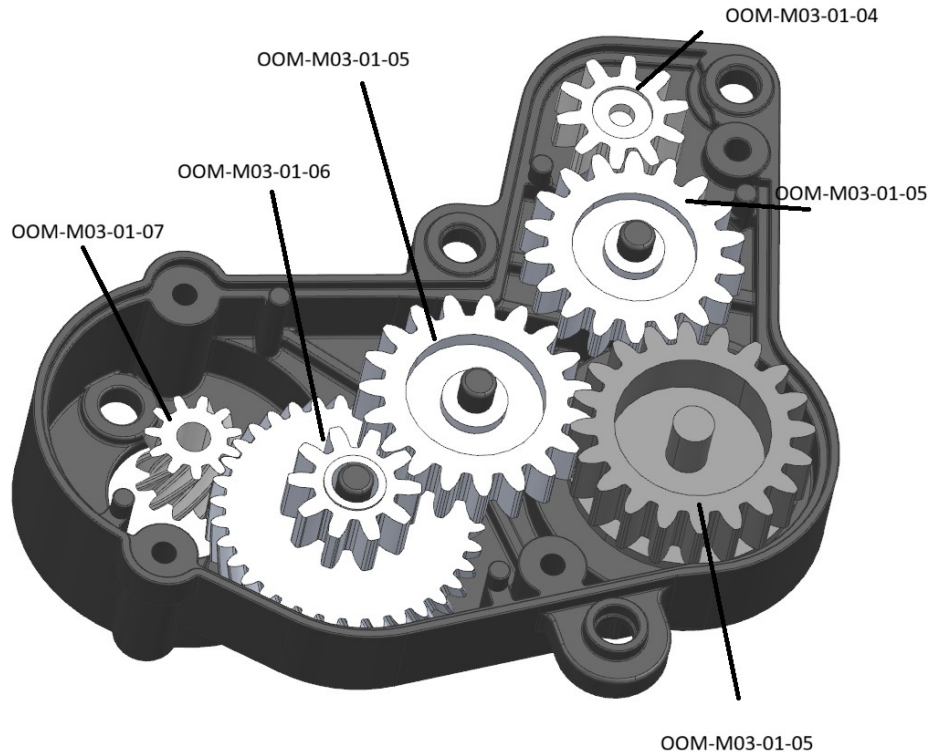


Figure 2. Internal OOM-M03-01 gearbox assembly with project part-code callouts.

Part code	Designation	Type	Primary role
OOM-M03-01-01	Bottom housing	Custom housing	Motor-side shell, lower gear-support features and structural mounting bosses.
OOM-M03-01-02	Top housing	Custom housing	Closes the gear cavity and carries the external output opening/interface.
OOM-M03-01-03	Output drive socket	Custom coupler	Non-circular external drive interface located on the Z7 output axis.
OOM-M03-01-04	Z7 output gear	10-tooth spur gear	Final internal gear; drives the external output socket axis.
OOM-M03-01-05	Intermediate gear set	Z4/Z5/Z6 spur positions	Annotated at the three intermediate spur-gear positions; tooth counts are 19T, 20T and 18T.
OOM-M03-01-06	Compound transfer gear	Z2 helical + Z3 spur	34T helical gear and 10T spur pinion rigidly share one axis.
OOM-M03-01-07	Z1 motor pinion	11-tooth helical pinion	Mounted on the RS-385PH-2466 motor shaft; gearbox input member.
RS-385PH-2466	DC motor	Purchased motor	12 VDC motor identification; drives Z1 directly.

PART-CODE READING / OOM-M03-01-05 is shown at the Z4, Z5 and Z6 positions. Their tooth counts are 19T, 20T and 18T respectively; use the CAD geometry for the exact position part.

2 Gear Train Design & Kinematics

The 3.091:1 reduction is created at the helical motor stage; the following four spur meshes route torque to the remote output axis without changing the net ratio.

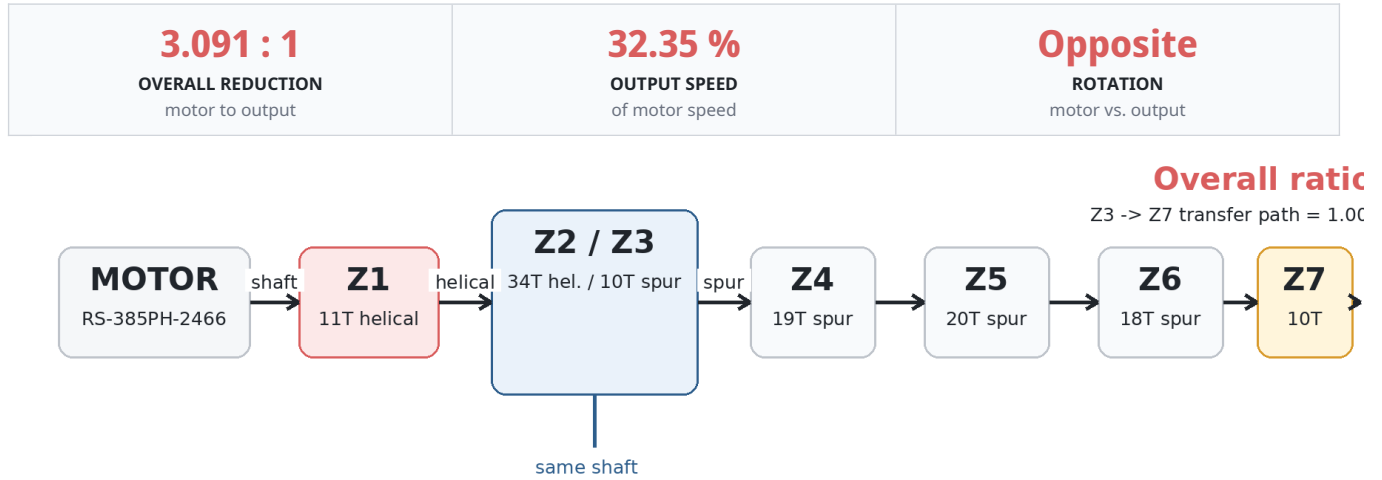


Figure 3. Kinematic chain. Z2 and Z3 are coaxial on the compound transfer gear.

Stage-by-stage ratio and center distance

Stage	Mesh	Teeth	Driven/driver ratio	Center distance	Function
Z1 -> Z2	Helical	11 -> 34	$34/11 = 3.0909$	12.413 mm	Primary reduction
Z2 / Z3	Compound axis	34 / 10	1.000	Same axis	Rigid coaxial transfer
Z3 -> Z4	Spur	10 -> 19	$19/10 = 1.9000$	11.600 mm	Transfer
Z4 -> Z5	Spur	19 -> 20	$20/19 = 1.0526$	15.600 mm	Transfer
Z5 -> Z6	Spur	20 -> 18	$18/20 = 0.9000$	15.200 mm	Transfer
Z6 -> Z7	Spur	18 -> 10	$10/18 = 0.5556$	11.200 mm	Transfer to output

Why the intermediate train is useful

The spur path from Z3 to Z7 telescopes to $Z7/Z3 = 10/10 = 1.000$. In other words, Z4, Z5 and Z6 mainly relocate the torque path through the housing and around the package geometry rather than adding net reduction. The total ideal ratio therefore becomes $(34/11) \times (10/10) = 3.0909$. Five external gear meshes are present in total, so the final output rotates opposite the motor shaft.

3 Gear Geometry Reference

Documented tooth parameters, derived helical-system relationships and involute reference values for CAD recreation or cross-checking.

Helical input stage - Z1 / Z2

Gear	z	Normal m [mm]	Pressure angle	Helix angle	Pitch d [mm]	Outside da	Root df	Derived db
Z1 / OOM-M03-01-07	11	0.500	25 deg	25 deg	6.069	7.069	4.819	5.397
Z2 / OOM-M03-01-06	34	0.500	25 deg	25 deg	18.757	19.757	17.507	16.679

HELICAL-SYSTEM INTERPRETATION / The documented pitch diameters satisfy $d = m_n z / \cos(\beta)$, so the 0.500 mm value behaves as a normal module. With $\beta = 25$ deg, the derived transverse module is $m_t = 0.5517$ mm. Treating the 25 deg pressure angle as normal gives a transverse pressure angle of approximately 27.23 deg. Preserve the native CAD helix hand and tooth form when remaking the pair.

Spur transfer gears - Z3 through Z7

Gear	z	Module [mm]	Pressure angle	Pitch d	Outside da	Root df	Derived db
Z3 / OOM-M03-01-06	10	0.800	25 deg	8.000	9.600	6.000	7.250
Z4 / OOM-M03-01-05	19	0.800	25 deg	15.200	16.800	13.200	13.776
Z5 / OOM-M03-01-05	20	0.800	25 deg	16.000	17.600	14.000	14.501
Z6 / OOM-M03-01-05	18	0.800	25 deg	14.400	16.000	12.400	13.051
Z7 / OOM-M03-01-04	10	0.800	25 deg	8.000	9.600	6.000	7.250

Reading the diameter data

- For the spur stages, the pitch diameters follow $d = m z$, the outside diameters follow $d_a = d + 2m$, and the documented root diameters correspond to $d_f = d - 2.5m$.
- For the helical pair, the same addendum/root offsets are applied using the 0.5 mm normal module while the pitch diameter is enlarged by $1/\cos(\beta)$.
- The involute base-circle values shown above are derived reference values ($d_b = d \cos(\alpha_t)$ for the helical pair and $d_b = d \cos(25 \text{ deg})$ for the spur gears). Use the native CAD/STEP tooth profile as the manufacturing master.

4 Top Housing & Output Interface

OOM-M03-01-02 closes the gear cavity and locates the external OOM-M03-01-03 output socket on the final gear axis.

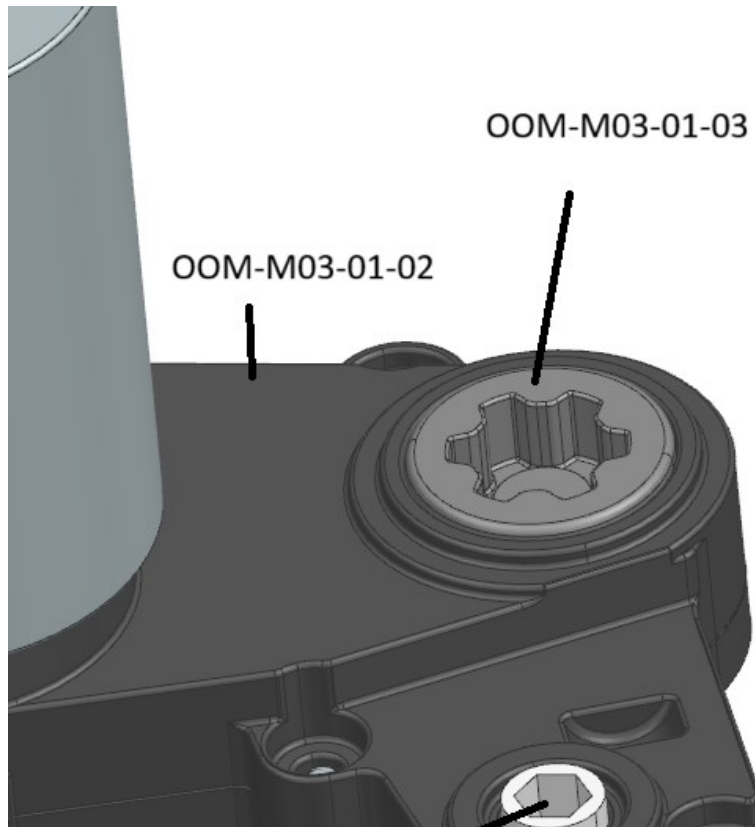


Figure 4. Top-housing region showing OOM-M03-01-02 and the OOM-M03-01-03 external output interface.

Feature	Type	Geometry / interface	Design role
OOM-M03-01-02	Top housing	Perimeter shell, closure ledge, bosses and upper support features	Maintains upper-side alignment of the gear axes and encloses the transmission.
OOM-M03-01-03	Output drive socket / coupler	Non-circular internal drive form visible in CAD	Transfers Z7 shaft torque to the external brush/actuator interface without relying on friction alone.
Z7 output axis	Gear + socket coaxial interface	10-tooth spur gear inside, output socket outside	The internal gear axis and external coupling axis must remain concentric through the closed housing.

Top-housing design logic

- The output opening is directly above the final gear axis, so torque exits without an offset coupler.
- The non-circular socket gives positive torque transfer. A remix may change this outer socket while preserving the Z7 axis and support geometry.
- The perimeter register and bosses preserve gear-axis spacing after closure and should not distort the output journal or remove end clearance.
- The top-housing support hardware is now identified separately: an AMC30604 metric plain Oilite bush and a CAD-defined Ø3.00 mm shaft/pin are shown in Section 6.2. Preserve their modeled locations and running clearances when remaking the cover.

5 Bottom Housing & Motor Interface

OOM-M03-01-01 carries the motor-side structure and establishes the alignment between the RS-385PH-2466 shaft and the helical Z1/Z2 mesh.

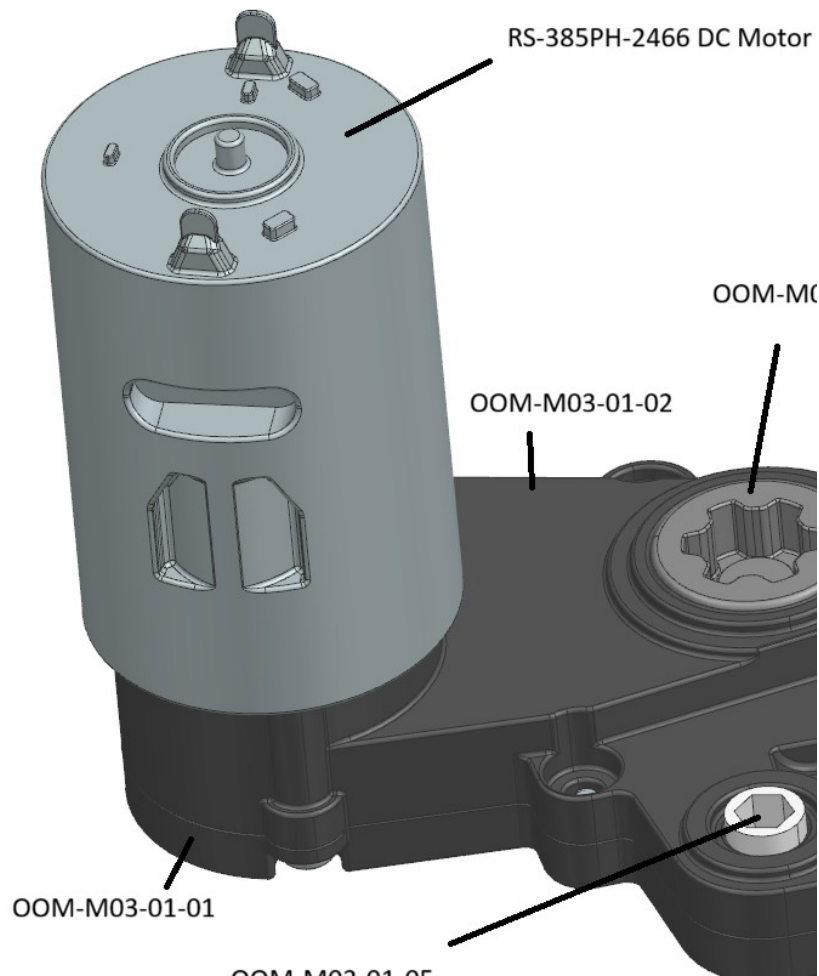


Figure 5. Motor-side view showing the RS-385PH-2466 motor and OOM-M03-01-01 / OOM-M03-01-02 housing interface.

Interface	Type	Reference	Design role
RS-385PH-2466	Brushed DC motor	12 VDC identification	Motor shaft drives Z1 directly; motor envelope and shaft location are defined by CAD.
OOM-M03-01-01	Bottom housing	Motor cradle / lower shell and gear cavity	Reacts motor body loads, supports lower-side gear axes and provides external mounting bosses.
OOM-M03-01-07 / Z1	11T helical pinion	0.5 mm normal module, 25 deg helix	Press/fit relationship to motor shaft defines the gearbox input.
Motor-to-Z2 alignment	Axis relationship	12.413 mm nominal center distance	Controls helical mesh contact, noise and radial/axial load transfer.

MOTOR-INTERFACE RULE / Preserve the CAD motor shaft location, Z1 axial position and 12.413 mm Z1-Z2 center distance. The helical first stage creates axial thrust, so the motor bearings and compound Z2/Z3 support must retain the two gears axially as well as radially.

6 Shaft, Bearing, Bushing & Load-Path Architecture

How the motor bearings, purchased plain bearings, fixed shafts/posts and housing seats carry the forces created by the seven-gear train.

Axis	Support concept	Rotating member	Load-path / design interpretation
Motor / Z1 axis	Motor internal bearings	OOM-M03-01-07 pinion on motor shaft	Radial load plus helical axial thrust. Keep pinion overhang short and preserve motor axial retention.
Z2 / Z3 compound axis	Ø2.45 mm fixed shaft/post in OOM-M03-01-01	OOM-M03-01-06 compound gear	The compound Z2/Z3 axis is located by a fixed Ø2.45 mm support shown in the bottom-housing reference. Preserve its CAD length, axis position and gear-bore running clearance.
Z4 axis	JSM-1416-05 iglidur J plain sleeve bearing	OOM-M03-01-05, 19T	14 mm ID × 16 mm OD × 5 mm long polymer sleeve. Provides a replaceable, lubricant-free plain-bearing interface at the large-hub transfer axis.
Z5 axis	Ø2.45 mm fixed shaft/post in OOM-M03-01-01	OOM-M03-01-05, 20T	Fixed gear axis. The 2.45 mm diameter is a design value; do not round it to 2.50 mm without rechecking the mating gear-bore clearance.
Z6 axis	Ø2.45 mm fixed shaft/post in OOM-M03-01-01	OOM-M03-01-05, 18T	Fixed gear axis feeding the final output stage. Preserve shaft perpendicularity and CAD-defined support length.
Z7 / output axis	MR106ZZ shielded miniature ball bearing	OOM-M03-01-04 + OOM-M03-01-03	6 mm ID × 10 mm OD × 3 mm wide rolling bearing at the final/output support location. Maintain seat concentricity and axial retention through housing closure.

Key geometric relationships

- Each gear-center distance is created by the housing journal/support locations. Print shrinkage or fit tuning must not move those axes independently.
- The Z2/Z3 compound gear is the kinematic hinge between the helical and spur systems. Any angular or radial play at this axis affects both the motor mesh and the entire downstream transfer path.
- Axial clearance is needed on every rotating gear so closing the housing does not create face rubbing. The helical Z2 member additionally needs a positive thrust reaction path.
- The three OOM-M03-01-05 positions should be checked individually because their tooth counts differ even though the annotated view groups them under one project code.
- Treat every support as a complete tribological interface: purchased bearing/bushing, shaft or hub diameter, housing seat, axial retention and clearance must remain compatible. Use the CAD for support lengths and axial stack-up, and the selected supplier catalogue for purchased-part tolerances.

LOAD-PATH SUMMARY / Motor torque enters through Z1, is reduced at Z1-Z2, passes coaxially from Z2 to Z3, then follows Z4-Z5-Z6 to Z7 and the output socket. The housings absorb the radial separating forces at every mesh, the helical thrust at the first stage, and the external output reaction through their bosses and mounting interfaces.

6.1 Bottom Housing Support Hardware

OOM-M03-01-01 carries three fixed Ø2.45 mm gear-support shafts/posts together with the MR106ZZ rolling bearing and the JSM-1416-05 polymer plain bearing. These parts convert the printed housing from a simple shell into the primary lower-side datum for the transmission axes.

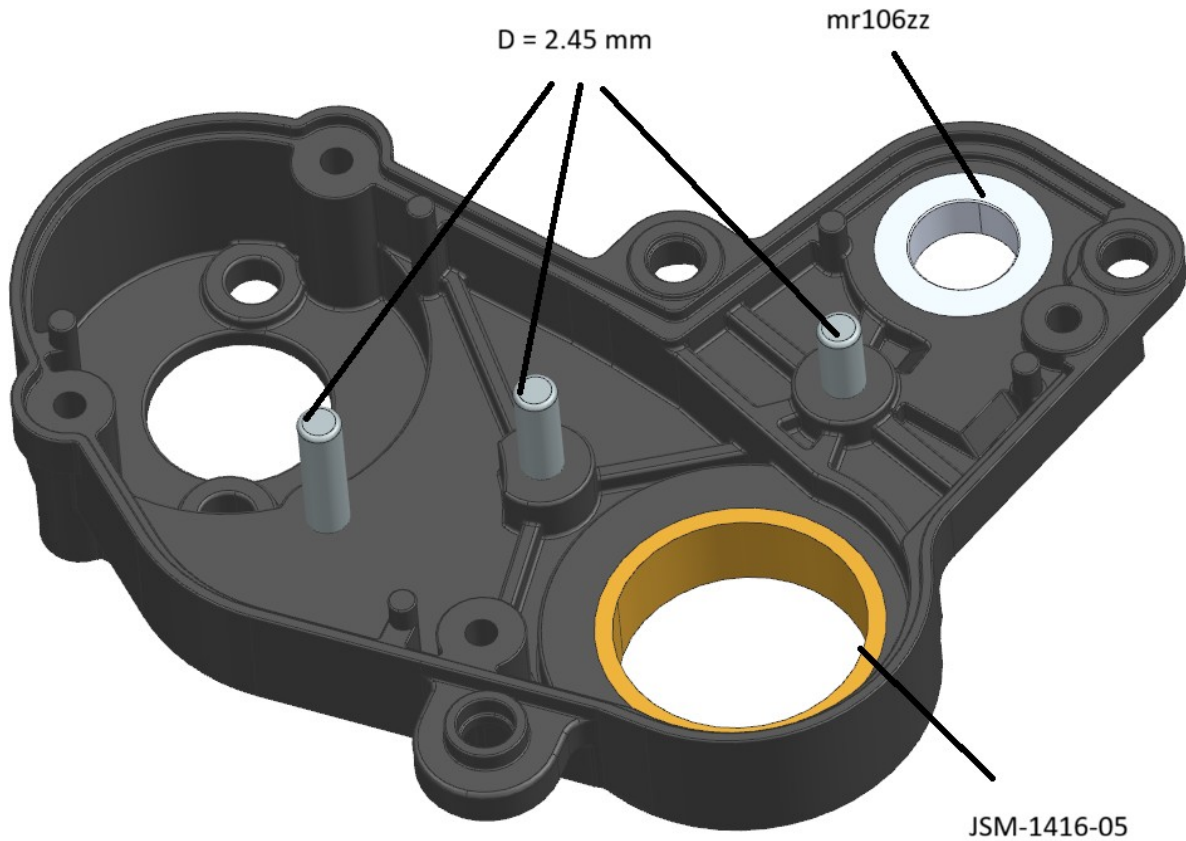


Figure 6. Bottom-housing support hardware: three Ø2.45 mm fixed shaft locations, MR106ZZ miniature ball bearing and JSM-1416-05 iglidur J sleeve bearing.

Component	Type	Nominal size	Qty	Design use
MR106ZZ	Shielded miniature deep-groove ball bearing	6 mm ID × 10 mm OD × 3 mm W	1	Rolling support at the CAD-defined final/output-side axis. Standard metal shields retain lubricant and limit debris entry.
JSM-1416-05	igus iglidur J cylindrical plain bearing	14 mm ID × 16 mm OD × 5 mm L	1	Large-diameter plain-bearing support. Lubricant-free polymer sleeve is press-fitted into the housing seat and runs on the mating 14 mm hub/shaft surface.
Ø2.45 mm shafts/posts	Fixed gear shafts / pins	Ø2.45 mm; length per CAD	3	Locate three internal rotating members. Preserve the exact diameter and perpendicularity; do not substitute Ø2.50 mm without checking gear-bore clearance.

CATALOG FIT NOTE / JSM-1416-05 is a 14 × 16 × 5 mm iglidur J sleeve. Manufacturer catalogues specify the finished running fit only after press-fit; printed housing variants should tune the bearing seat with a local fit coupon rather than scaling the complete housing.

6.2 Top Housing Support Hardware

The inside face of OOM-M03-01-02 adds a compact Oilite sleeve-bearing location and a separate Ø3.00 mm shaft/pin. Together with the bottom-housing supports, these features close the axial stack and provide replaceable wear surfaces where direct plastic-on-shaft contact would otherwise dominate.

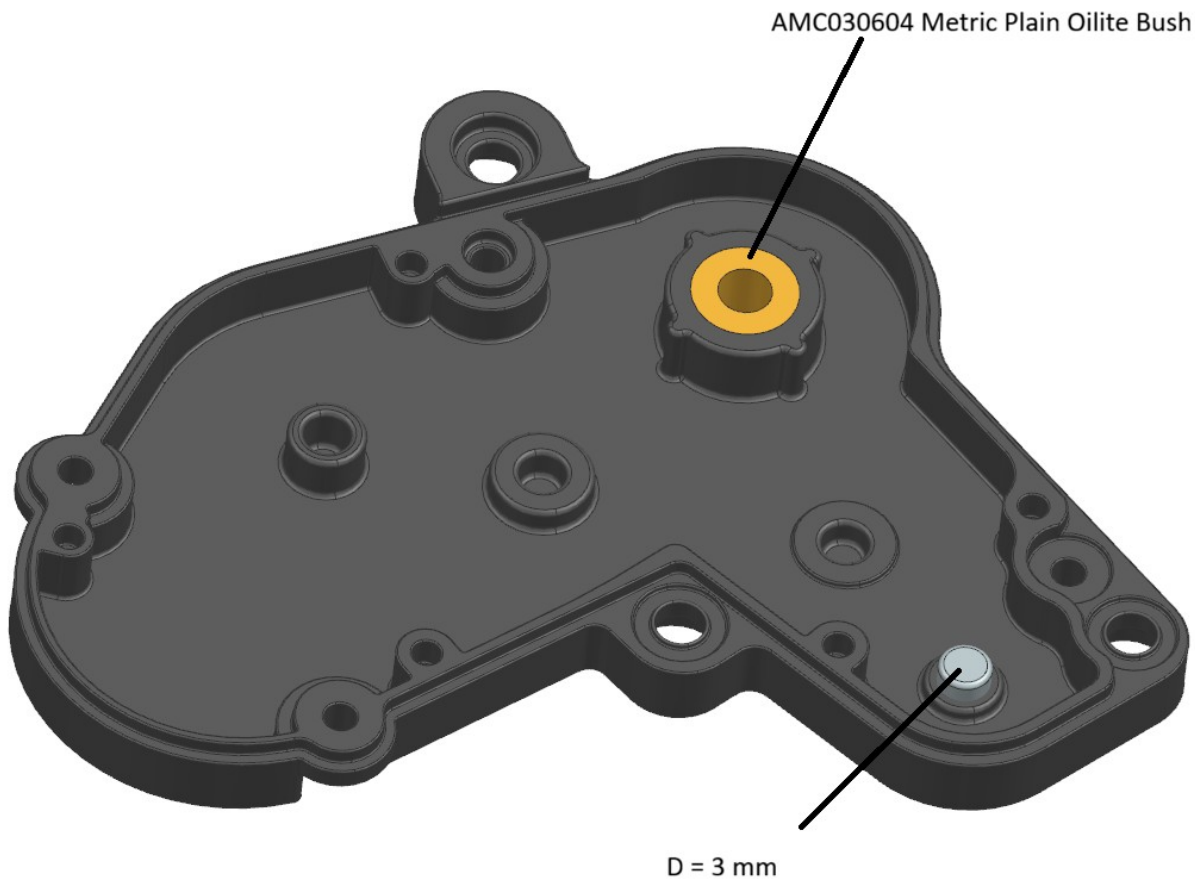


Figure 7. Top-housing support hardware: AMC30604 metric plain Oilite bush and CAD-defined Ø3.00 mm shaft/pin location.

Component	Type	Nominal size	Qty	Design use
AMC30604	Metric plain Oilite bush	Nominal 3 mm ID × 6 mm OD × 4 mm L	1	Self-lubricating sintered-bronze sleeve at the annotated top-housing support boss. Supplier codes may vary; preserve the 3 × 6 × 4 mm geometry and Oilite/plain-bearing function.
Ø3.00 mm shaft/pin	Fixed shaft / pin	Ø3.00 mm; length per CAD	1	Mating support shaft shown in the top housing. Axial length and end condition are CAD-controlled; confirm running clearance against the selected bushing supplier tolerance.

Support-interface rules

- Do not treat bearing OD, bushing OD and shaft diameter as independent dimensions: the housing seat, press fit, running fit and axial retention form one tolerance chain.
- The annotated CAD calls the bronze sleeve AMC30604. A 3 × 6 × 4 mm metric Oilite sleeve is a standard catalogue geometry, but supplier-specific part numbers and tolerance systems can differ.
- Shaft lengths and materials are not specified by the annotated support views; use the current CAD/BOM for those values rather than inferring them from the diameter callouts.

7 BOM, Assembly & Manufacturing Notes

Core parts and a practical build sequence for community reproductions and CAD remixes.

Item	Part no.	Description	Qty	Origin	Design detail
1	OOM-M03-01-01	Bottom housing	1	Custom CAD	Motor-side shell and lower support features
2	OOM-M03-01-02	Top housing	1	Custom CAD	Upper shell and output-interface region
3	OOM-M03-01-03	Output drive socket / coupler	1	Custom CAD	External positive-drive interface on Z7 axis
4	OOM-M03-01-04	Z7 final output gear, 10T	1	Custom CAD	m 0.8 spur; final internal gear
5	OOM-M03-01-05	Intermediate spur gear positions Z4/Z5/Z6	3 positions	Custom CAD	19T / 20T / 18T; use exact CAD per position
6	OOM-M03-01-06	Compound Z2/Z3 transfer gear	1	Custom CAD	34T helical + 10T spur on one axis
7	OOM-M03-01-07	Z1 motor pinion, 11T	1	Custom CAD	m_n 0.5; beta 25 deg; motor shaft input
8	RS-385PH-2466	Brushed DC motor	1	Purchased	12 VDC identification; CAD defines fit
9	MR106ZZ	Miniature shielded ball bearing	1	Purchased	6 × 10 × 3 mm; bottom-housing rolling support
10	JSM-1416-05	iglidur J cylindrical plain bearing	1	Purchased	14 × 16 × 5 mm; lubricant-free polymer sleeve
11	AMC30604	Metric plain Oilite bush	1	Purchased	Nominal 3 × 6 × 4 mm; top-housing sleeve bearing
12	CAD-defined Ø2.45 shafts	Fixed gear shafts / pins	3	Cut to CAD length	Ø2.45 mm; preserve diameter and perpendicularity
13	CAD-defined Ø3.00 shaft	Fixed shaft / pin	1	Cut to CAD length	Ø3.00 mm; length/end condition per CAD

Suggested assembly sequence

1. Inspect the housing journal/post surfaces, gear bores and motor pocket; remove only intentional support material or debris from functional surfaces.
2. If the Z1 pinion is installed separately, fit OOM-M03-01-07 to the RS-385PH-2466 shaft using the CAD axial position and a method that does not hammer load through the motor bearings.
3. Seat the motor in OOM-M03-01-01 and place the OOM-M03-01-06 compound Z2/Z3 gear so the helical teeth enter mesh without forcing.
4. Install Z4, Z5 and Z6 in their CAD-defined positions, then install Z7 and engage the OOM-M03-01-03 output socket/shaft relationship as modeled.
5. Close OOM-M03-01-02 lightly and rotate the train by hand through several revolutions. Expect motor cogging, but no hard gear interference, housing rub or tight spots.
6. Tighten the housing evenly using the current CAD/BOM fastener strategy, repeat the rotation check, then begin powered run-in at reduced voltage before full-load use.

Manufacturing and modification notes

- Preserve the helical gear hand, 25 deg helix angle, normal module and compound Z2/Z3 coaxial relationship when moving the design between CAD systems.
- Do not assume the three OOM-M03-01-05 gear positions are geometrically identical; their tooth counts and center-distance roles differ.
- If journal fits are printed, tune them with process-specific coupons rather than globally scaling the entire housing.
- Record material, print process, orientation, post-cure and dimensional compensation for gears and housings so another builder can reproduce the same behavior.
- Use only a small compatible gear lubricant if lubrication is required; keep it away from motor commutator openings and any plain-bearing material that should run dry.

8 Community Build Notes & References

A compact checklist for downloading the CAD, reproducing the gearbox or adapting the motor/output interfaces.

What to look for after assembly

Check	What the builder should observe
Motor alignment	Z1 should enter the Z2 helical mesh evenly across the intended face width; no visible skew or end contact.
Helical mesh	Rotate slowly and confirm no hard spot. The first stage carries both radial force and axial thrust.
Compound axis	Z2 and Z3 should run as one coaxial member without wobble between the helical and spur stages.
Transfer path	Z4-Z5-Z6 should rotate freely and preserve positive backlash around the full revolution.
Output axis	OOM-M03-01-03 should remain concentric with Z7 and rotate without rubbing the top housing.
Housing closure	Fastening the two shells should not significantly increase rotation torque or create gear-face rub.
Run-in	Start at reduced voltage, record sound/current, then inspect teeth, journals and housing witness marks.
Remix documentation	Record any changed motor, shaft, journal, gear or socket dimensions so the modified version is reproducible.

Design reference measurements

Characteristic	Design reference	Note
Z1-Z2 center distance	12.413 mm	Helical stage
Z3-Z4 center distance	11.600 mm	Spur stage
Z4-Z5 center distance	15.600 mm	Spur stage
Z5-Z6 center distance	15.200 mm	Spur stage
Z6-Z7 center distance	11.200 mm	Spur stage
Overall ideal ratio	3.0909 : 1	Output speed = 32.35% of motor speed

References and project links

- [1] OOMWOO One CAD repository - <https://github.com/makerspet/oomwoo-one-cad>
- [2] OOMWOO One CAD 3D design specification - <https://github.com/makerspet/oomwoo-one-cad/blob/main/docs/SPEC.md>
- [3] OOMWOO main project BOM and design documentation - <https://github.com/makerspet/oomwoo>
- [4] Mabuchi Motor RS-385PH family product information - <https://product.mabuchi-motor.com/>
- [5] IEC 60335 appliance test documentation identifying RS-385PH-2466 as a 12 VDC motor (Roborock appliance component list).
- [6] SMB Bearings - MR106ZZ miniature shielded radial ball bearing data sheet; 6 mm bore × 10 mm OD × 3 mm width.
- [7] igus - iglidur J JSM-1416-05 cylindrical plain bearing; 14 mm ID × 16 mm OD × 5 mm length.
- [8] Oilite - M-Series metric 3 × 6 × 4 mm sleeve-bearing catalogue geometry (supplier-specific part numbering varies; annotated CAD identifier: AMC30604).

REPOSITORY NOTE / For current license terms, contribution guidance, part revisions and manufacturing geometry, use the repository README, LICENSE and native CAD files. This PDF is an explanatory design reference, not a substitute for the current CAD revision.